Quiz 3 — Sample 1

DATA 412/612

Weeks 1–13. The new material is R Markdown presentations, chunk options, tables, and citations.

Write your answers in an R script or a short `.Rmd`. You may use R and RStudio. You may use your notes and the course website (`https://amousavi1.github.io/612/data-612.html`). You may not use Google, any other website, or an LLM. You do not have to knit. Write the YAML, the chunk headers, and the code you would put in the file.

1. Write a minimal YAML header for an ioslides presentation titled `Delay notes`, with you as the author. Under it, write the markdown for two slides after the title slide. Each slide has a title and one sentence. Say which heading level starts a slide in ioslides.

2. You need three chunks:

   - a setup chunk that loads the tidyverse and is not printed;
   - a chunk that shows both the code and a `knitr::kable()` table of `mpg` summarized to the median `hwy` by `class`, with the caption `Median highway mileage`;
   - a plot chunk whose code is hidden, whose messages are hidden, and whose figure still appears.

   Write the chunk options for each. Set the shared defaults in the setup chunk with

   `knitr::opts_chunk$set()`.

   Override them only where a chunk differs.

3. Point the document at a bibliography file `refs.bib`. In the body, cite the key `wickham2017r4ds` in parentheses, and end the file with a `References` heading so the entry is printed.

4. Still in that document, add a chunk with `echo = TRUE` that runs a two-sample `t.test()` of `hwy` for `drv "f"` against `drv "r"` in `mpg`. State the null hypothesis in one sentence under the chunk.